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Hexagulation numbers: symmetry breaking on the gyroid surface

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The regular arrangements of identical spheres on the gyroid surface have been investigated using Monte Carlo simulations of hard spheres undergoing the Alder transition. In our preceding study, we showed that these regular structures on saddle-shaped triply periodic minimal surfaces are systematically classified by the hexagulation number H, analogous to the Caspar–Klug triangulation number for icosahedral viruses. In this paper, we focus on the non-centrosymmetric regular structures that emerge when H is a multiple of three on the gyroid surface. In these cases, the space group $Ia\bar{3}d$ of the gyroid is reduced to its non-centrosymmetric subgroup $I\bar{4}3d$. By considering both inversion structures, we construct an efficient bilayer configuration that forms a truss-like structure. These physically efficient designs may provide inspiration for future applications in soft-matter science, DNA-origami nanotechnology, metamaterials, and even architectural design.

This article is part of the theme issue 'Geometry, materials and the imagination'.

1. Introduction

The gyroid surface discovered by Schoen appears ubiquitously in nature and materials, ranging from butterfly wings to the self-assembly of lipids, surfactants and polymers [1,2]. Owing to its unique geometry, the gyroid has been studied extensively across mathematics, physics, chemistry and biology [3–9]. In the past 5 years in particular, research on gyroid structures has expanded rapidly in both scope and depth. Recent studies span geometry and topology [10,11], polymers [12–17], surfactants [18–20], photonics [21,22], phononics [23], magnetism [24], electronic properties [25], materials science [26], superconductivity [27], proton-conducting surfactants [28], metal–organic frameworks [29], molecular simulations [30,31], biological photonic structures [32], cell biology [33,34] and even bird evolution [35]. Interest in gyroid-based architectures has also grown in engineering applications such as architecture and three-dimensional printing [36].

In this paper, we extend our study of regular particle arrangements on gyroid surfaces. On a flat surface, spherical particles generically form a hexagonal lattice, a universal structure arising from dense packing, space partitioning, entropic effects or interparticle interactions. This naturally raises the question of how regular particle arrangements are modified when the underlying surface is curved. On spherical surfaces, this problem dates back to the classical work of J. J. Thomson [37] and was later examined by Fejes Tóth [38] and Mackay and co-workers [39]. Goldberg [40] identified families of polyhedra composed of pentagonal and hexagonal faces, whose face centres correspond to particle positions, and Caspar & Klug subsequently developed a construction principle for icosahedral virus capsids inspired by geodesic domes [41,42]. More recently, self-assembled Goldberg polyhedra have been synthesized experimentally [43].

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By contrast, regular particle arrangements on saddle-shaped surfaces with negative Gaussian curvature remain far less explored [44–47]. Mackay & Terrones [48] proposed periodic tiling on a primitive surface inspired by fullerene networks. Related tilings and embeddings have been studied mathematically [49,50], and many three-dimensional networks have been enumerated [51,52]. Hierarchical and multiscale structures on gyroid surfaces have also been investigated [53,54]. In particular, Pedersen & Hyde [55] presented more than 80 distinct embeddings of two-dimensional hyperbolic honeycombs into three-dimensional Euclidean space, yielding triply periodic polyhedra with cubic symmetry. Among these are 10 infinite deltahedra composed entirely of equilateral triangles, which have recently been realized experimentally via DNA origami self-assembly [56–58]. One notable example is 'gyrangles', a previously little-known infinite regular polyhedron that is topologically equivalent to the gyroid surface [58,59].

It is well known that even a system of purely hard spheres exhibits a fluid–solid transition, the Alder transition, leading to hexagonal ordering [60]. This entropy-driven ordering mechanism is now recognized as a universal feature of soft-matter systems [61]. Above a critical density, the ordered phase gains more entropy than the disordered fluid. An important open question is whether analogous regular arrangements emerge on triply periodic minimal surfaces. Motivated by this issue and by the discovery of complex hierarchical gyroid structures [62], we initiated a systematic investigation of hard spheres confined to minimal surfaces. On the gyroid surface, we indeed observed Alder-like transitions, but only for specific numbers of spheres [63]. To clarify the origin of these special cases and establish a unified framework, we extended our Monte Carlo simulations to the primitive and diamond surfaces. This led to the identification of so-called 'hexagulation numbers' $H = h^2 + k^2 - hk$, in analogy with the Caspar–Klug triangulation numbers $T = h^2 + k^2 + hk$ for icosahedral viruses, where h and k are nonnegative integers [64]. A detailed explanation of the hexagulation number is provided in the supplementary material, notes SI.

In the present work, we focus on the gyroid surface and extend our previous results to $H = 21$. We examine not only single gyroid membranes but also double gyroid layers, motivated by the construction of saddle-shaped truss frameworks. A hexagonal arrangement corresponds to cross sections of the hexagonal close-packed (hcp) and face-centred cubic (fcc) lattices along their threefold axes, and the fcc hard-sphere crystal is stabilized purely by entropy. When the membrane consists of two coupled layers of spheres, a natural question arises: what is the most efficient saddle-shaped truss structure compatible with the gyroid geometry? We explore this question under the assumption that double layers on the gyroid surface can form close-packed, truss-like frameworks.

The remainder of the paper is organized as follows. Section 2 describes the Monte Carlo simulation method. Section 3 presents the results for single- and double-gyroid membranes. Section 4 summarizes our conclusions.

2. Material and methods

We have performed Monte Carlo (MC) simulations of hard spheres whose centres are confined to a rigid gyroid membrane. The gyroid surface is described by the commonly used approximate form [65,66] $g_0(\mathbf{r}) = 0$, where

$$g_0(\mathbf{r}) = \sin x \cos y + \sin y \cos z + \sin z \cos x, \tag{2.1}$$

with $\mathbf{r} = (x, y, z)$. The lattice constant is set to unity, and the factors of 2π are omitted throughout. For single-membrane systems, the simulation protocol follows that is described in [63,64]. In this case, the system is treated in the NVT ensemble, where N is the number of hard spheres and V is the volume defined by $|g_0(\mathbf{r})| < 0.1$. The detailed balance is fulfilled. The temperature T has no physical meaning in this simulation because the system is athermal.

For double-membrane systems, two gyroid sheets separated by a half-width $w > 0.1$ are introduced via

$$g_\pm(\mathbf{r}) = \sin x \cos y + \sin y \cos z + \sin z \cos x \pm w. \tag{2.2}$$

The parameter w does not correspond to the exact normal distance between the two surfaces; instead, it serves as a control parameter that varies approximately linearly with their mean separation. Half of the sphere centres are constrained to the region $|g_+(\mathbf{r})| < 0.1$, while the remaining half are confined within $|g_-(\mathbf{r})| < 0.1$, where $g_\pm(\mathbf{r})$ is defined in equation (2.2). Since $w > 0.1$, the two membranes do not overlap. Consequently, no sphere can move from the region $|g_+(\mathbf{r})| < 0.1$ to $|g_-(\mathbf{r})| < 0.1$, or vice versa.

Simulations are initialized from random (fluid-like) configurations with sufficiently small sphere radii on each gyroid surface. MC simulations are then carried out without imposing any symmetry constraints. A typical MC cycle consists of the following steps: (i) a sphere is selected at random; (ii) a trial displacement is generated from uniform distributions within a rectangular box of size Δx, Δy and $\Delta z = \pm 0.0075$; (iii) the trial move is rejected if $|g_\pm(\mathbf{r})| > 0.1$ or if any hard-sphere overlap occurs; otherwise, it is accepted; (iv) every 20 cycles, an update of the half-width w is attempted, as described below; (v) steps (i)–(iv) are repeated; and (vi) the acceptance ratio and the instantaneous value of w are recorded. At each sphere radius, 10^7 MC steps are typically performed. The sphere radius is then increased (or decreased) by +0.0001 (−0.0001), and the procedure is repeated.

When the half-width w is fixed, the system samples the NVT ensemble, as in the single-membrane case, and a detailed balance is satisfied for sphere movement. However, the optimal value of w is not known a priori for each particle diameter. Therefore, the half-width w is treated as an adaptive variable to facilitate efficient exploration of configurations and is controlled by an auxiliary force term F through the factor $\exp(-Fw)$, which acts as a solvent-like bias.

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Keeping the sphere positions fixed, with a probability of 0.5, one of the following updates is attempted:
$w \to w - \Delta w$, accepted with probability 1,
$w \to w + \Delta w$, accepted with probability $\exp(-F\Delta w)$.

Movements that would lead to hard contacts between spheres and the membranes are forbidden. For small values of F, the half-width increases and the two membranes effectively separate, whereas for large F, the membranes collapse owing to strong confinement. Unless otherwise stated, we use $F = 100$ and $\Delta w = 0.0001$.

It should be emphasized that the updates of w are introduced as a numerical device to enhance the sampling efficiency and are not intended to define a true equilibrium ensemble. Because the volume changes with w, a detailed balance is not strictly satisfied for the combined move set including w-updates, and the system does not correspond to a well-defined *NVT* or *NPT* ensemble. Moreover, the hard-sphere pressure P inside the membrane volume differs from the auxiliary force F, further indicating that the system is not equivalent to an *NPT* ensemble.

Nevertheless, we have verified that the resulting structures are robust against variations in the frequency of w-updates (e.g. every 10–40 cycles), and no qualitative change in the stable structures was observed. This finding indicates that the equilibrium-like states obtained are not sensitive to the specific update schedule. For efficiency, we adopt an update every 20 cycles.

Distances between spheres are computed using the three-dimensional Euclidean metric rather than the geodesic distance along the surface. Accordingly, the particles are modelled as hard spheres, not hard disks.

3. Results

3.1. Single membrane

We first consider a single membrane defined by $g_0(\mathbf{r})$. As a representative example, we present the results for the case $H = 12$ with $N = 192$ spheres. Figure 1a shows the acceptance ratio (AR) of the MC sphere movements as a function of the sphere diameter d. As expected, the AR decreases monotonically with increasing d because the available free volume for the sphere movement is reduced. Superimposed on this overall trend, a pronounced jump is observed near $d \approx 0.122$, signalling a phase transition associated with an entropy gain upon ordering.

Figure 1b displays the order parameter $h(d)$ as a function of d, defined by

$$h(d) = \frac{1}{N}\sum_{i=0}^{N} f_{422}\left(\mathbf{r}_i\right), \tag{3.1}$$

where the sum runs over all sphere centres $\mathbf{r}_i = (x_i, y_i, z_i)$ and $\langle\cdots\rangle$ denotes the MC average. The function $f_{422}(\mathbf{r})$ is invariant under the symmetry operations of the space group $I\bar{4}3d$ appropriate to the gyroid surface [67], and is given by

$$\begin{aligned} f_{422}\left(\mathbf{r}_i\right) = {} & \sin(2x)\sin\left(2y\right)\sin(4z) \\ & + \sin(2x)\sin\left(4y\right)\sin(2z) \\ & + \sin(4x)\sin\left(2y\right)\sin(2z)\,. \end{aligned} \tag{3.2}$$

In the fluid phase, $h(d) = 0$, whereas in the solid phase $h(d)$ acquires a finite positive or negative value depending on spontaneous symmetry breaking during the simulation.

The transition points identified from the AR (figure 1a) and from the order parameter (figure 1b) coincide within numerical accuracy. This agreement indicates that, near the transition, the ordered phase possesses more available free volume for particle motion than the disordered fluid phase, and therefore a higher entropy. The transition is clearly first order. Owing to the three-dimensionally connected nature of the gyroid surface, the transition is significantly sharper than that observed in ordinary two-dimensional flat systems. Figure 1c visualizes a pair of inversion-related structures, corresponding to the left ($h(d) > 0$) and right ($h(d) < 0$) arrangements associated with positive and negative values of the order parameter, respectively.

Our simulations show that clear fluid–solid transitions occur only for specific numbers of spheres per unit cell. These special cases are classified by the hexagulation number H defined in §1. Figure 2 summarizes the ordered phases with $N_{\mathrm{reg}} = 48$, 64, 112, 144, 192, 208, 256, 304 and 336, corresponding to $H = 3$, 4, 7, 9, 12, 13, 16, 19 and 21, respectively. The cases $H = 3$, 4, 7 and 9 were reported previously. We find that the transition diameter is approximately represented by $d_{\mathrm{T}} \cong \frac{0.422}{\sqrt{H}}$. For values of N away from N_{reg}, the AR and order parameter vary smoothly with d, resembling glass-like behaviour. For values of N close to N_{reg}, transitions are sometimes observed, but the resulting structures typically contain point defects. When H is a multiple of three, as in the present example, the centrosymmetric space group $Ia\bar{3}d$ is broken to the non-centrosymmetric space group $I\bar{4}3d$. This symmetry reduction reflects the absence of spheres on the threefold rotational centres for such values of H. The two structures are related by the inversion transformation $(x, y, z) \to (1 - x, 1 - y, 1 - z)$. Note that $I\bar{4}3d$ is distinct from the non-centrosymmetric but chiral subgroup $I4_132$ known for the single gyroid phase observed in ABC triblock copolymers.

The supplementary material, table S1 summarizes the regular tessellations of hard spheres on the gyroid surface, classified by the hexagulation number H, space group, Wyckoff positions and polygonal tiling. The supplementary material, table S2 lists the sphere coordinates obtained from the MC simulations. Using the Wyckoff positions tabulated in the international tables for

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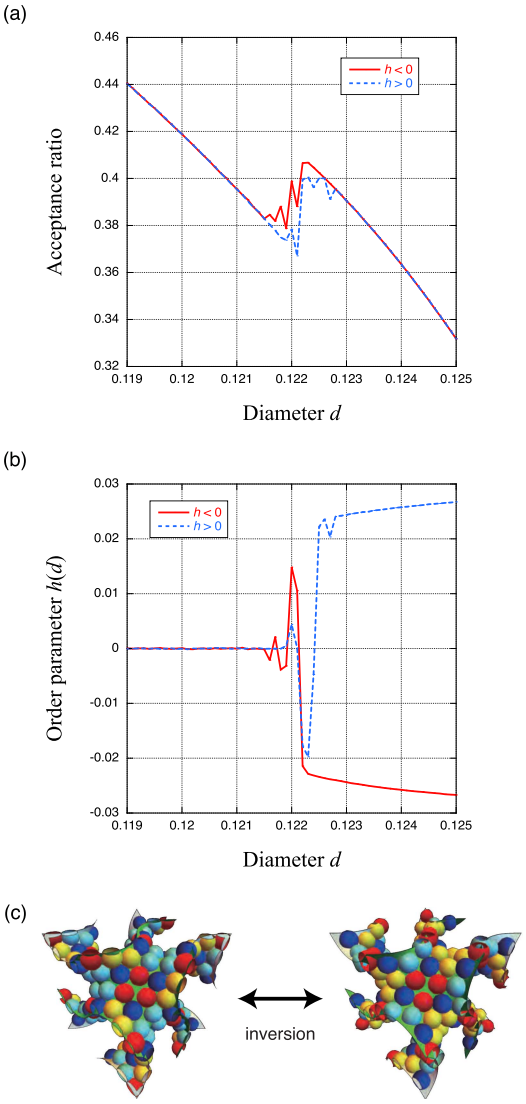


Figure 1. (a) Acceptance ratio (AR) as a function of sphere diameter for the $H = 12$, $N = 192$ system. (b) Plots of $h(d)$ for two independent MC simulations, showing both $h < 0$ and $h > 0$ after transitions. (c) Two versions of regular tessellations on the gyroid surface, which give negative (left) and positive (right) order parameters, respectively. Different colours indicate Wyckoff positions 48e. Their coordinates are related by the inversion transformation $(x, y, z) \rightarrow (1 - x, 1 - y, 1 - z)$.

crystallography, all the coordinates can be reconstructed. For visualization purposes, it is also instructive to project the sphere positions onto the Poincaré disk representation of hyperbolic geometry [54], since a dodecagonal region of the disk can be conformally mapped onto the gyroid surface and covers half of the surface area within a cubic unit cell.

3.2. Double membrane

We next consider the double-membrane system and focus on the representative case $H = 12$, as shown in figures 3 and 4. Figure 3a presents the AR as a function of the sphere diameter d. A clear jump is observed slightly below $d = 0.12$, indicating the Alder transition. The transition diameter is lower than that of the corresponding single-membrane system, because spheres in the two layers make contact.

As shown in figure 3b, the half-width w exhibits a clear jump at the transition point, indicating a volume expansion upon solidification. Such behaviour is rather unprecedented. This jump becomes evident only when the MC simulation is run for sufficiently long cycles. When the MC cycles are too short, decreases in w—corresponding to the collapse of two layers—are observed instead, and no expansion occurs. This behaviour is more intuitive, since increased structural regularity would normally lead to a more compact volume. The emergence of volume expansion upon crystallization is intriguing. A detailed investigation of its origin is left for future work.

In figure 4a, a snapshot of the red and blue spheres is shown together with the gyroidal surfaces $|g_+(\mathbf{r})| = 0.1$ and $|g_-(\mathbf{r})| = 0.1$. As discussed in §1 as a working hypothesis, figure 4b shows a view along a threefold axis. In this projection, one can clearly see that the red and blue spheres alternate along the stacking direction. Specifically, the red spheres form a triangular lattice, and half of the interstitial sites in this triangular arrangement are occupied by the blue spheres in the adjacent layer beneath. This results in a close-packed structure similar to fcc- or hcp-like stacking. This alternating occupation is significant because it supports

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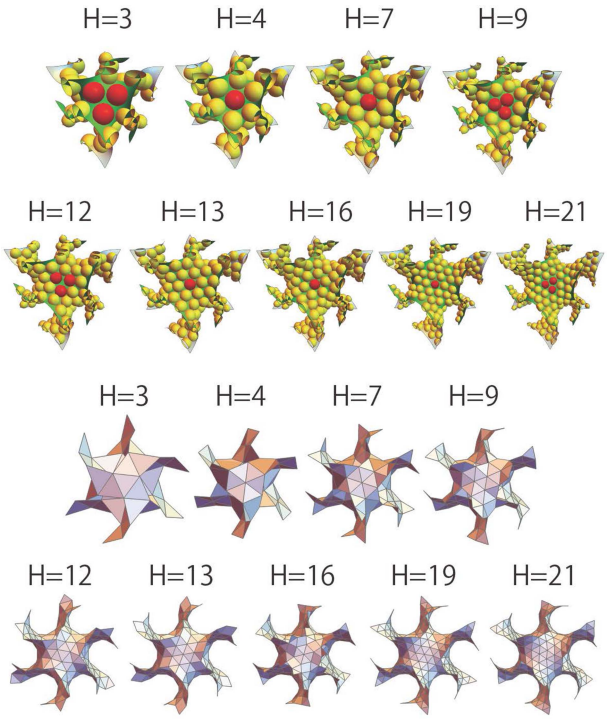


Figure 2. Regular structures of equal spheres on the gyroid surface with polygonal tiling connecting sphere centres. H is the hexagulation number: $H = 3, 4, 7, 9, 12, 13, 16, 19$ and 21. The corresponding number of vertices (spheres) per cubic unit cell is $N_{reg} (= 16H) = 48, 64, 112, 144, 192, 208, 256, 304$ and 336, respectively.

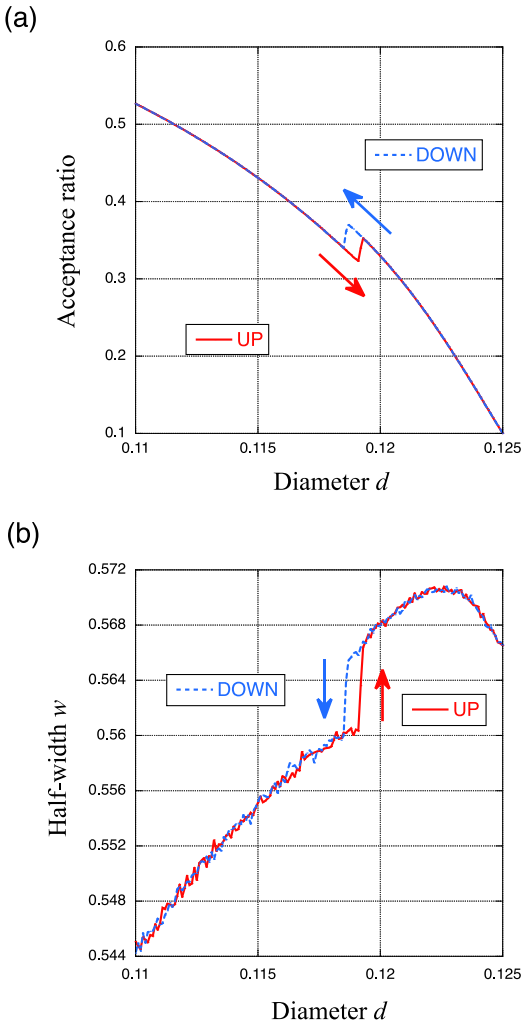


Figure 3. Double-membrane system. (a) Acceptance ratio (AR) as a function of sphere diameter for the $H = 12$ system with $n = 192$ spheres in each layer. (b) Half-width w between the two membranes, automatically determined by the auxiliary force $F = 100$.

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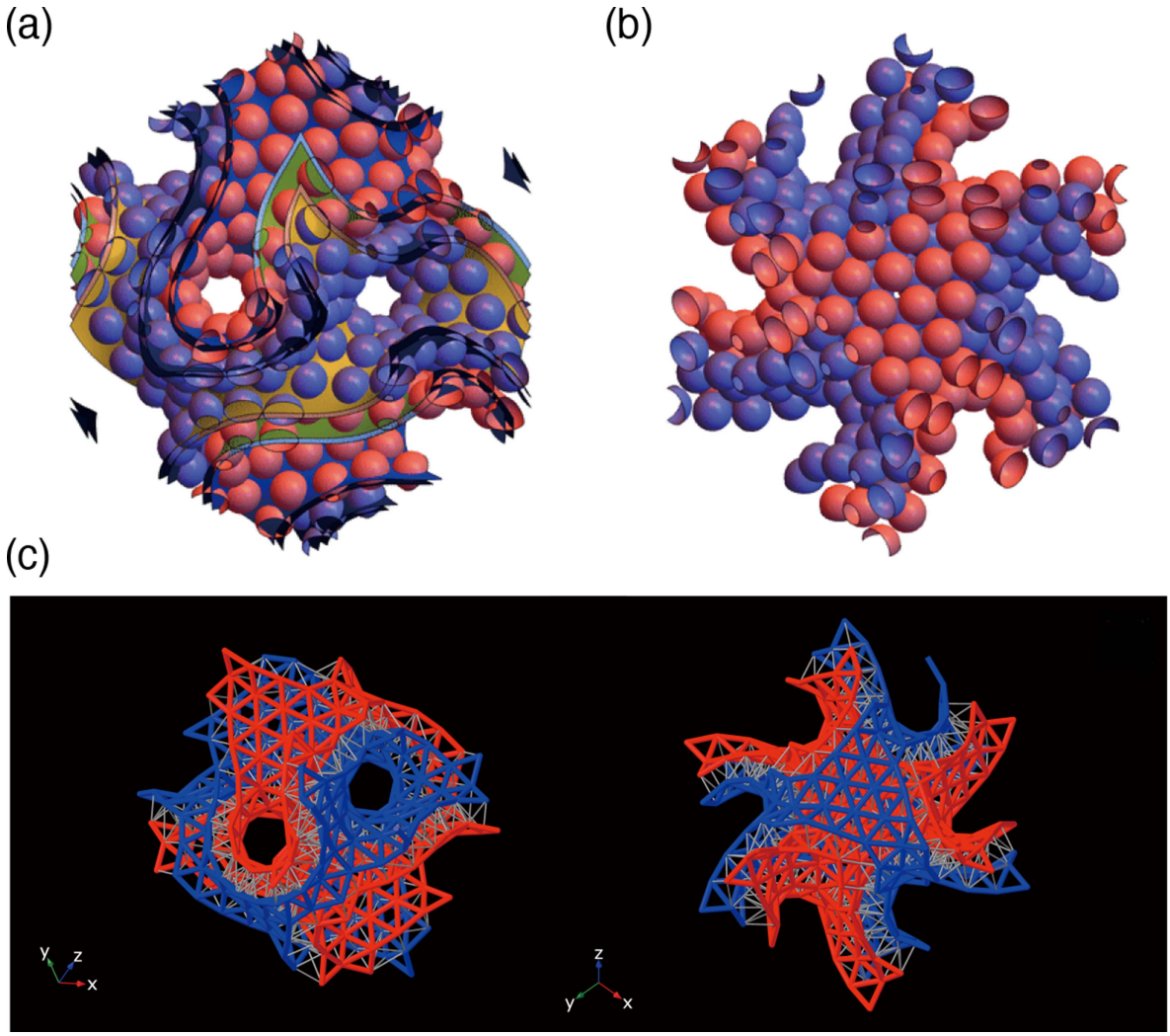


Figure 4. Structure of the double-membrane system for the $H = 12$ system with $N = 192$ spheres in each layer at $d = 0.125$. Half-width w between the two membranes, automatically determined by the auxiliary force $F = 100$. (a) Rendering of the red and blue spheres together with the four gyroidal surfaces $|g_+(\mathbf{r})| = 0.1$ and $|g_-(\mathbf{r})| = 0.1$. (b) A view along a threefold direction. In the centre region, the triangular arrangement is clearly observed. (c) Two views of the truss structures formed by the sphere centres. Struts connecting vertices in the same layer are coloured red or blue, while those connecting vertices in different layers are shown in white.

the proposed structural model and explains the observed packing symmetry. Finally, figure 4c presents two views of the truss structures formed by the sphere centres.

4. Summary

Hard-core repulsion is one of the most fundamental interactions in soft-matter science, where a wide variety of gyroid phases appear. In this work, we extended our study of how identical hard spheres can arrange themselves regularly on a gyroid minimal surface by performing MC simulations across the Alder transition. We show that our classification scheme, which is based on the hexagulation number H, remains valid even for large systems containing up to 336 particles per unit cell.

Our main finding is that despite the geometric complexity of the saddle-shaped gyroid surface, only a limited number of physically efficient arrangements are possible. In this sense, the situation resembles the restricted and highly symmetric structures observed in icosahedral viruses. Such regular sphere packings on curved surfaces may inspire future applications, including structured soft materials such as bijels, colloidosomes, and polymersomes [68–71], as well as DNA-origami-based self-assembled systems [72].

When H is a multiple of three, we further identify a bilayer structure that forms a truss-like network composed of inversion layers. Designing and understanding the mechanical properties of such rigid truss networks—capable of forming free-standing three-dimensional gyroid architectures—remains an interesting open challenge, with potential applications in metamaterials and architectural design.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. Data and codes are available within the manuscript and/or supporting information. Supplementary material: details of the hexagulation number are provided in supplementary material, notes SI. Supplementary material, notes SII includes tables S1 and S2 for regular

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tessellations of hard spheres on the G-surface. Data for the single- and double-membrane systems shown in figures 1a,b and 3a,b are provided as supplementary material, results 1 and 2, respectively, and their codes are provided as supplementary material, codes S1 and S2, respectively.

Supplementary material is available online [73].

Declaration of AI use. The authors used generative AI (ChatGPT, OpenAI) for language editing and grammar improvement. The authors take full responsibility for the content of this manuscript.

Authors' contributions. T.D.: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, software, supervision, validation, visualization, writing—original draft; K.Y.: data curation, formal analysis, investigation, visualization.

Both authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests

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